

Claim Safe

Auto Insurance premiums recommendation system



Abstract

- ▶ Our Auto Insurance Premium Recommendation System dynamically calculates personalized insurance premiums using advanced algorithms that assess risk based on driver history, vehicle details, and historical claims data. This system integrates multiple data sources, including Carfax reports, to ensure accurate and equitable premium suggestions aligned with insurer's strategic goals.

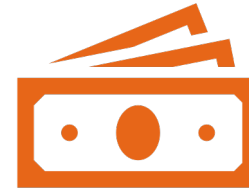
Overview



Risk Assessment Model: Analyzes driver history, vehicle information, and past claims to assess risk levels.



Data Integration: Utilizes diverse data sources such as Carfax reports, driving records, and vehicle databases to ensure comprehensive risk profiling.



Dynamic Pricing Algorithm: Adjusts premiums based on insurance company's strategic goals, reflecting current risk assessments of the drivers and market conditions.

Risk Assessment Model



Comprehensive Risk Profiling: Integrates driver history, vehicle details, and claims data to assess risk.



Uses data from Carfax reports, DMV records, and insurance claim histories.



Dynamic Data Analysis: Evaluates risk factors such as driver age, driving violations, and accident history.



Considers vehicle specifics including make, model, age, mileage, and safety features.

Input for Risk Assessment

- ▶ Driver's data
- ▶ Driver's payment history
- ▶ Driver's claims history
- ▶ Driver's vehicle Carfax reports



Driver's Data

```
{
  "driver_information": {
    "name": "John Doe",
    "address": "123 Main St, Anytown, VA, 12345",
    "date_of_birth": "1985-06-15",
    "driver_license_number": "VA123456789",
    "license_class": "Class D"
  },
  "license_status": {
    "status": "Active",
    "expiration_date": "2029-06-15"
  },
  "violations": [
    {
      "date": "2020-05-10",
      "violation": "Speeding 10-14 mph above the posted speed limit",
      "points": 4,
      "duration": "5 years"
    },
    {
      "date": "2021-03-15",
      "violation": "Failure to obey highway sign",
      "points": 3,
      "duration": "3 years"
    }
  ],
  "accidents": [
    {
      "date": "2019-07-08",
      "type": "Minor",
      "description": "Rear-end collision, minor damage",
      "at_fault": "Yes",
      "points": 3,
      "duration": "3 years"
    }
  ]
},
```

```
  "history": {
    "suspensions": [
      {
        "date": "2018-01-01",
        "reason": "Driving under the influence",
        "reinstated_date": "2018-07-01",
        "points": 6,
        "duration": "11 years"
      }
    ],
    "safe_driving_points": {
      "total_safe_points": 2
    }
  },
  "vehicle_information": {
    "vin": "1HGCM82633A004352",
    "make": "Honda",
    "model": "Civic",
    "year": 2018,
    "vehicle_use": "Personal",
    "annual_mileage": 15000,
    "safety_features": [
      "Automatic Emergency Braking",
      "Electronic Stability Control",
      "Adaptive Headlights"
    ]
  },
  "insured_for": 3,
  "location": {
    "type": "Urban",
    "population_density": "High",
    "crime_rate": "Moderate"
  }
}
```

Payment history Data

Claims history Data

```
{  
  "driver_license_number": "VA123456789",  
  "ssn": "XXX-XX-8164",  
  "credit_score": 750,  
  "monthly_payments_on_time": true  
},
```

```
{  
  "driver_license_number": "VA123456789",  
  "previous_claims": {  
    "minor_claims": 1,  
    "total_loss": 1  
  }  
}
```

Carfax report

```
{
  "carfax_reports": [
    {
      "vin": "1HGCM82633A004352",
      "make": "Honda",
      "model": "Civic",
      "year": 2018,
      "title_status": "Clean",
      "ownership_history": {
        "number_of_owners": 1,
        "purchase_type": "Personal"
      },
      "accident_history": {
        "minor_accidents": 3,
        "major_accidents": 0
      },
      "service_history": {
        "service_records": 3
      },
      "repair_history": {
        "minor_repairs": 2,
        "major_repairs": 1
      },
      "open_recalls": {
        "number_of_open_recalls": 1
      },
      "current_mileage": {
        "date": "2024-04-26",
        "mileage": 50000
      }
    }
  ],
}
```

Car Market Values

```
"Car Models": {
  "Toyota Camry": {
    "Year": 2019,
    "Market Value": 21000,
    "Repair Costs": {
      "Minor": 450,
      "Major": 3900
    }
  },
  "Honda Accord": {
    "Year": 2018,
    "Market Value": 19500,
    "Repair Costs": {
      "Minor": 440,
      "Major": 3800
    }
  },
  "Ford F-150": {
    "Year": 2020,
    "Market Value": 26000,
    "Repair Costs": {
      "Minor": 480,
      "Major": 4100
    }
  }
},
```

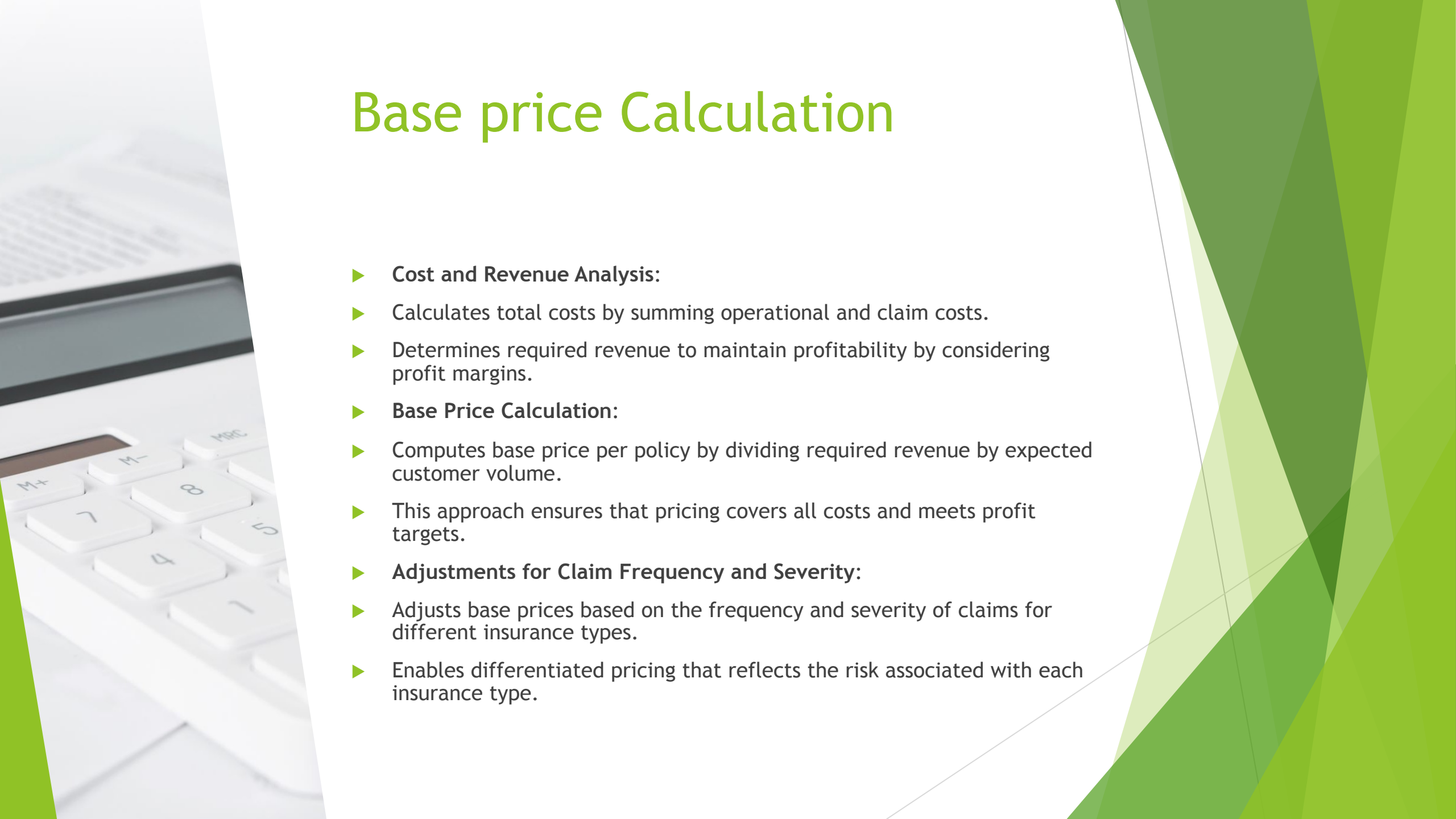



Factors considered for risk assessment

- ▶ **Driving History Analysis:** Evaluates patterns and frequency of traffic violations, accidents where the driver was at fault, and license suspensions to gauge driver behavior and propensity for risk.
- ▶ **Credit Score Influence:** Incorporates the driver's credit score as an indicator of financial responsibility, where higher scores may imply more responsible behavior, potentially correlating with safer driving habits.
- ▶ **Payment Timeliness:** Assesses the regularity of the driver's insurance premium payments. On-time payments are a sign of reliability, whereas late payments increase the risk score.
- ▶ **Claims History:** Reviews past claims to identify patterns in minor and major claims, adjusting the risk based on the frequency and severity of past claims, reflecting the driver's historical claim behavior.
- ▶ **Vehicle Condition and Age:** Factors in the vehicle's age and mileage, considering older or heavily-used vehicles as higher risk due to potential for more frequent breakdowns and higher maintenance costs.
- ▶ **Carfax Reports:** Utilizes Carfax reports to check for any past major repairs, accidents, or issues like a rebuilt title, which significantly increase the vehicle's risk profile.
- ▶ **Location Risk:** Takes into account the vehicle's garaging location, evaluating urban, suburban, or rural settings and local crime rates to adjust risk based on potential for theft, vandalism, and accident rates in the area.
- ▶ **Safety Features:** Credits vehicles equipped with advanced safety features like automatic emergency braking and stability control, which can mitigate the severity of or prevent accidents, thereby reducing risk.

Risk Model output

```
{
  "driver_information": {
    "name": "John Doe",
    "address": "123 Main St, Anytown, VA, 12345",
    "date_of_birth": "1985-06-15",
    "driver_license_number": "VA123456789",
    "license_class": "Class D"
  },
  "license_status": {
    "status": "Active",
    "expiration_date": "2029-06-15"
  },
  "vehicle_information": {
    "vin": "1HGCM82633A004352",
    "make": "Honda",
    "model": "Civic",
    "year": 2018,
    "vehicle_use": "Personal",
    "annual_mileage": 15000,
    "safety_features": [
      "Automatic Emergency Braking",
      "Electronic Stability Control",
      "Adaptive Headlights"
    ]
  },
  "risk_score": 15.36,
  "risk_level": "High Risk"
},
```



Base price Calculation

- ▶ **Cost and Revenue Analysis:**
 - ▶ Calculates total costs by summing operational and claim costs.
 - ▶ Determines required revenue to maintain profitability by considering profit margins.
- ▶ **Base Price Calculation:**
 - ▶ Computes base price per policy by dividing required revenue by expected customer volume.
 - ▶ This approach ensures that pricing covers all costs and meets profit targets.
- ▶ **Adjustments for Claim Frequency and Severity:**
 - ▶ Adjusts base prices based on the frequency and severity of claims for different insurance types.
 - ▶ Enables differentiated pricing that reflects the risk associated with each insurance type.

Input For Base price Calculation

```
{
  "company_info": {
    "name": "InsuraMax",
    "address": {
      "street": "123 poplar st",
      "city": "Jersey City",
      "state": "NJ",
      "zip_code": "23675"
    },
    "phone_number": "555-1234",
    "email": "contact@insuramax.com",
    "website": "www.insuramax.com"
  },
  "services": [
    {
      "type": "Liability Insurance",
      "description": "Covers damages to others caused by the policyholder while operating an automobile."
    },
    {
      "type": "Collision+Liability Insurance",
      "description": "Covers damage to the policyholder's car from a collision."
    },
    {
      "type": "liability+collision+Comprehensive Insurance",
      "description": "Covers damage to the policyholder's car that is not caused by a collision."
    },
    {
      "type": "Roadside Assistance",
      "description": "Provides assistance such as towing, battery jump start, flat tire change, and lock-out assistance."
    }
  ]
},
```

```
  "operational_costs": 350000,
  "claim_costs": 170000,
  "profit_margin": 0.10,
  "expected_customer_volume": 1000,
  "claim_frequency_and_severity": {
    "liability": {
      "frequency": 0.20,
      "severity": 800
    },
    "liability+collision": {
      "frequency": 0.15,
      "severity": 3000
    },
    "liability+collision+comprehensive": {
      "frequency": 0.12,
      "severity": 8000
    }
  }
}
```



Optimal Premium Evaluation

- ▶ **Integration of Vehicle Details and Risk Factors:**
 - ▶ Incorporates vehicle-specific data like market value and major repair costs from a centralized car database.
 - ▶ Factors in driver-specific data such as safe driving points and suspension history.
- ▶ **Dynamic Discount Application:**
 - ▶ Utilizes the all function from the dgalPy library to evaluate complex conditions for applying discounts.
 - ▶ Offers discounts based on risk score, annual mileage, and the number of safety features, promoting safer driving behaviors.
- ▶ **Calculation of Customized Premiums:**
 - ▶ Calculates premiums based on risk levels, applying different multipliers for varying risk scores.
 - ▶ Adjusts premiums dynamically, applying additional factors based on vehicle market value and repair costs for high-risk vehicles.

```
{
  "name": "John Doe",
  "address": "123 Main St, Anytown, VA, 12345",
  "date_of_birth": "1985-06-15",
  "driver_license_number": "VA123456789",
  "license_class": "Class D",
  "vehicle_information": {
    "vin": "1HGCM82633A004352",
    "make": "Honda",
    "model": "Civic",
    "year": 2018,
    "vehicle_use": "Personal",
    "annual_mileage": 15000,
    "safety_features": [
      "Automatic Emergency Braking",
      "Electronic Stability Control",
      "Adaptive Headlights"
    ]
  },
  "risk_score": 15.36,
  "premiums": {
    "premiums": {
      "liability": "$92.22",
      "liability+collision": "$128.47",
      "liability+collision+comprehensive": "$192.22"
    },
    "discount_applied": false
  }
},
```

```
{
  "name": "Angela Thompson",
  "address": "789 Willow Ave, Arlington, VA, 78901",
  "date_of_birth": "1984-04-22",
  "driver_license_number": "VA667788990",
  "license_class": "Class D",
  "vehicle_information": {
    "vin": "1HGCV1F13JA004321",
    "make": "Honda",
    "model": "Accord",
    "year": 2018,
    "vehicle_use": "Personal",
    "annual_mileage": 5000,
    "safety_features": [
      "Automatic Emergency Braking",
      "Electronic Stability Control",
      "Adaptive Headlights"
    ]
  },
  "risk_score": 5.66,
  "premiums": {
    "premiums": {
      "liability": "$60.87",
      "liability+collision": "$84.79",
      "liability+collision+comprehensive": "$126.87"
    },
    "discount_applied": true
  }
},
```

```
{
  "name": "Jane Smith",
  "address": "234 Elm St, Richmond, VA, 23456",
  "date_of_birth": "1992-08-20",
  "driver_license_number": "VA987654321",
  "license_class": "Class D",
  "vehicle_information": {
    "vin": "2T1BURHE8JC015872",
    "make": "Toyota",
    "model": "Corolla",
    "year": 2020,
    "vehicle_use": "Personal",
    "annual_mileage": 10000,
    "safety_features": [
      "Automatic Emergency Braking",
      "Electronic Stability Control",
      "Adaptive Headlights"
    ]
  },
  "risk_score": 13.62,
  "premiums": {
    "premiums": {
      "liability": "$75.93",
      "liability+collision": "$105.78",
      "liability+collision+comprehensive": "$158.26"
    },
    "discount_applied": true
  }
}
```

Outputs

THANK YOU

